A PROJECT REPORT

on

"A Framework for Classifying Kidney Stones Using Explainable Transformer"

Submitted to

KIIT Deemed to be University

In Partial Fulfilment of the Requirement for the Award of

BACHELOR'S DEGREE IN

COMPUTER SCIENCE ENGINEERING

BY

Satwik Gupta	2305075
Saurabh Sharma	2305076
Shirsh Mohan	2305245
Soham Dey	2305250
Swayam Kashyap	2305258

UNDER THE GUIDANCE OF

Prof. Roshini Pradhan

SCHOOL OF COMPUTER ENGINEERING

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

BHUBANESWAR, ODISHA - 751024

KIIT Deemed to be University

School of Computer Engineering
Bhubaneswar, ODISHA 751024

CERTIFICATE

This is certify that the project entitled

"A Framework for Classifying Kidney Stones Using Explainable Transformer"

submitted by

Satwik Gupta	2305075
Saurabh Sharma	2305076
Shirsh Mohan	2305245
Soham Dey	2305250
Swayam Kashyap	2305258

is a record of bona fide work carried out by them, in the partial fulfilment of the requirement for the award of the Degree of Bachelor of Engineering (Computer Science & Engineering OR Information Technology) at KIIT Deemed to be University, Bhubaneswar. This work is done during the year 2025-2026, under our guidance.

Date: / /

\`

Prof. Roshini Pradhan
Project Guide

Acknowledgements

We are profoundly grateful to **Prof. Roshini Pradhan** of **School of Computer Engineering, KIIT University** for her expert guidance, insightful suggestions, and continuous encouragement throughout the entire project. Her deep expertise, constructive feedback, and constant motivation played a crucial role in shaping our ideas and refining our approach. From the initial stages of conceptualization to the final execution, she consistently ensured that our work remained focused and aligned with its objectives. Her mentorship not only helped us overcome challenges but also inspired us to strive for excellence, ultimately enabling this project to successfully achieve its intended goals.

i

Satwik Gupta

Signature of GROUP MEMBER

Saurabh Sharma

Signature of GROUP MEMBER

Shirsh Mohan

Signature of GROUP MEMBER

Soham Dey

Signature of GROUP MEMBER

Swayam Kashyap

Signature of GROUP MEMBER

ABSTRACT

Renal failure poses a significant public health challenge, and the global shortage of nephrologists has motivated the development of **AI-driven systems** for **automatic diagnosis of kidney disorders**. This work targets three primary renal disease categories kidney stones, cysts, and tumors and leverages **12,446 annotated CT images** of the abdomen and urograms to construct an AI-based diagnostic framework and support research toward a digital twin of renal function. **Exploratory analysis** revealed that all classes share similar mean color distributions, making simple statistical discrimination inadequate and necessitating more powerful representation-learning methods. To address this, six models were developed: three based on advanced **Vision Transformer architectures (EANet, DINO, and Swin Transformer)** and three based on established deep learning backbones **ResNet50**, **VGG16**, and **Vision Transformer (ViT)** with modified final layers.

Among all evaluated models, **ViT** achieved the highest accuracy of **99.37%**, and comparisons of F1 score, precision, and recall demonstrated that **transformer-based architectures outperform conventional CNN** models for multi-class kidney disease classification. Grad-CAM analysis further showed that **DINO** and **VGG16** localize anatomical abnormalities more precisely than **ResNet50**, which exhibited comparatively weaker performance. The combination of **ViT**'s exceptional accuracy and **DINO**'s interpretability indicates strong potential for deploying these models as decision-support tools in diagnosing kidney tumors, cysts, and stones in clinical settings.

Keywords: Kidney Stone Classification, CT medical imaging, Explainable AI (Grad-CAM), Vision Transformer (ViT), Multi-class kidney disease detection

Contents

Chapter 1
Introduction

Kidney disease has become an important public health concern because chronic kidney disease affects more than 10 percent of the global population, and delayed diagnosis of renal abnormalities can lead to severe complications including renal failure. The study focuses on four CT-based categories relevant to renal diagnosis: normal kidney, cyst, stone, and tumor. CT imaging is especially suitable for this problem because it offers high-resolution cross-sectional anatomical information that supports reliable identification of kidney abnormalities.

The project addresses the shortage of nephrologists and radiologists by proposing an AI-driven diagnostic framework that can automatically classify kidney conditions from CT scans. Existing studies often rely on ultrasound images or limited binary classification settings, which leaves a gap for robust multi-class CT-based diagnosis. This report is therefore organized around the problem definition, methodology, implementation flow, evaluation plan, standards adopted, and the project conclusion with future scope.

Chapter 2

Basic Concepts/ Literature Review

Kidney disease is a major global health concern, with chronic kidney disease affecting more than ten percent of the world's population and projected to become one of the top causes of mortality in the coming decades. Among renal disorders, cysts, stones, and tumors are particularly important because they directly affect kidney function and often require timely intervention. Clinicians commonly use imaging modalities such as X-ray, ultrasound, MRI, and especially CT scans along with pathology tests to diagnose these conditions.

Despite these tools, diagnosis remains challenging due to limited specialist availability and the subtle appearance of early-stage abnormalities in medical images. Traditional approaches and basic statistical image features are often insufficient to distinguish between closely related classes such as cysts, tumors, and stones, whose pixel-intensity distributions largely overlap. This has motivated the use of modern deep learning and transformer-based methods that can model complex spatial and contextual patterns in CT images and support accurate automated classification.

2.1 Kidney Diseases and CT Imaging

Kidney disorders such as cysts, stone disease (nephrolithiasis), and renal cell carcinoma represent common yet serious clinical problems. Simple kidney cysts are fluid-filled sacs bounded by a thin membrane, typically exhibiting low Hounsfield unit values on CT, whereas stones are crystalline deposits that can obstruct urinary flow and cause acute symptoms. Renal cell carcinoma is a malignant tumor and ranks among the ten most widespread cancers worldwide, making reliable imaging-based detection crucial for patient outcomes.

CT imaging is widely preferred for detailed kidney assessment because it produces high-resolution cross-sectional slices that reveal fine anatomical structure across the abdomen and urinary tract. Both contrast-enhanced and non-contrast CT protocols can be used to visualize cysts, stones, tumors, and normal tissue, enabling more precise localization than ultrasound in many cases. However, when these images are assessed manually, subtle differences between classes and large dataset sizes can make diagnosis time-consuming and subject to inter-observer variation, which motivates the use of automated AI-based analysis.

2.2 Deep Learning and Vision Transformers

Deep learning has achieved strong results in medical image classification and segmentation by automatically learning hierarchical features from raw pixels. Convolutional neural networks such as VGG16, ResNet, EfficientNet, and Xception are widely used because they extract local patterns like edges and textures, and transfer learning from ImageNet allows effective training even when annotated medical data are limited. Nevertheless, purely CNN-based models can struggle to capture long-range dependencies due to their localized receptive fields, limiting their ability to represent global anatomical context in complex CT volumes.

Vision Transformers (ViT) and related transformer architectures address this limitation by treating images as sequences of patches and applying self-attention to model global relationships across the entire image. Variants such as Swin Transformer and EANet improve efficiency and multi-scale representation, while self-supervised methods like DINO learn rich visual features without requiring extensive manual labels. When combined with explainability tools such as Grad-CAM, these transformer-based models not only achieve high accuracy—such as the reported 99.37 percent accuracy of ViT on the CT kidney dataset—but also provide visual explanations that highlight clinically relevant regions, increasing trust in AI-assisted diagnosis.

Chapter 3

Problem Statement / Requirement Specifications

Manual diagnosis of kidney abnormalities from CT scans requires expert interpretation and can be time-consuming, especially in regions where specialists are limited. Existing studies often focus on binary classification or on lower-detail modalities like ultrasound, which may not be sufficient for accurate multi-class renal disease detection.

The problem addressed in this project is the development of a robust AI system that can automatically classify CT images into normal, stone, cyst, and tumor categories with high accuracy and explainability. The system should support early diagnosis, reduce diagnostic workload, and provide interpretable results suitable for real-world medical use.

3.1 Project Planning

The project begins with problem identification and literature study on kidney disease diagnosis using AI and medical imaging. After reviewing prior work, the next phases include dataset collection, annotation, preprocessing, model selection, training, evaluation, and interpretability analysis.

A practical project schedule includes requirement analysis, system design, implementation, model training, testing, result interpretation, and final documentation. This phased plan ensures that each module is developed and validated before deployment or report submission.

3.2 Project Analysis (SRS)

The Software Requirement Specification for this project includes both functional and non-functional requirements. Functionally, the system must accept CT image inputs, preprocess them, classify them into renal condition categories, and display interpretable predictions using visual explanations.

Non-functional requirements include high classification accuracy, reliability, scalability for larger datasets, and usability for medical support environments. The model should also provide fast inference and support reproducible evaluation using metrics such as accuracy, precision, recall, F1 score, and confusion matrix.

3.3 System Design

The system is designed as a machine learning pipeline that starts with CT image acquisition and ends with prediction and interpretation. Major components include data input, preprocessing, feature learning model, classification module, evaluation module, and explainability module.

The design follows a modular approach so that CNN-based and transformer-based models can be trained and compared within the same experimental workflow. This allows flexible benchmarking and easier future enhancements.

3.3.1 Design Constraints

The main design constraints include dataset size, computational cost of transformer models, requirement for annotated CT scans, and domain sensitivity of medical diagnosis. Model interpretability is also a constraint because clinical applications demand decisions that can be visually justified rather than opaque predictions.

Another limitation is that real-world deployment would require validated clinical datasets, ethical approvals, and rigorous testing beyond experimental performance. Therefore, the current system should be treated as a decision-support model rather than a standalone diagnostic authority.

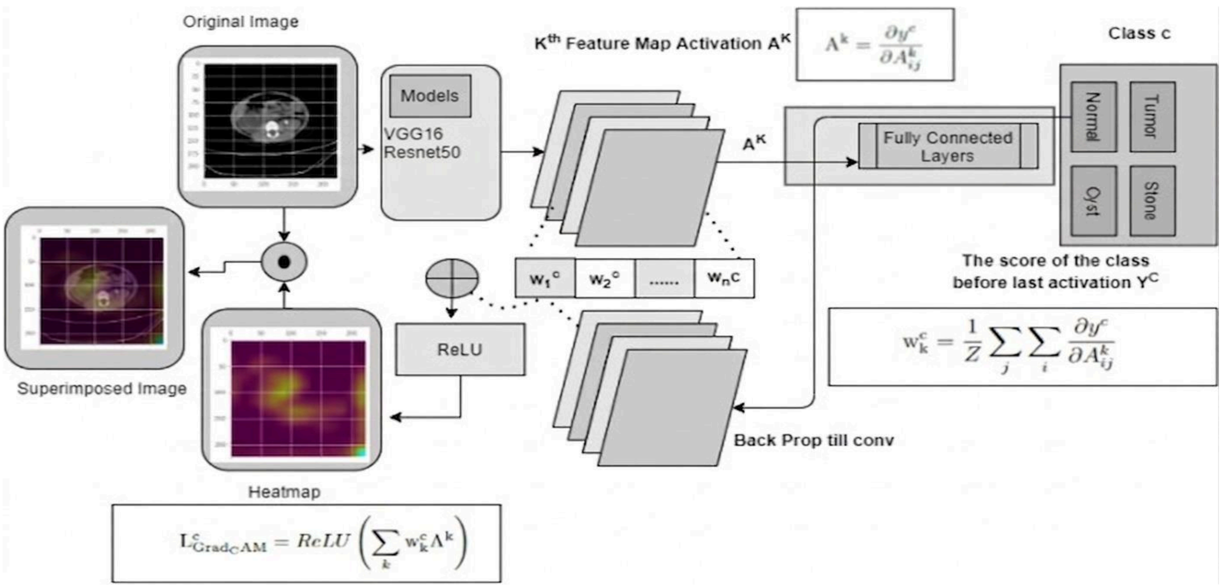
3.3.2 System Architecture / Block Diagram

The block-level architecture of the proposed system can be described as follows:

- CT Image Dataset Input
- Image Preprocessing and Normalization
- Training/Test Split
- Deep Learning and Transformer Models
- Prediction Layer for Normal, Cyst, Stone, and Tumor Classes
- Performance Evaluation Module
- Grad-CAM / Explainability Visualization
- Final Diagnostic Support Output

A simple workflow representation is:

Input CT Images → Preprocessing → Feature Extraction / Model Training → Classification → Evaluation → Explainability Output



Chapter 4

Implementation

The implementation uses a dataset of 12,446 annotated CT images covering normal kidneys, cysts, tumors, and stones. Six models are considered in the study, including VGG16, ResNet50, Vision Transformer (ViT), EANet, Swin Transformer, and DINO.

The development workflow includes dataset preparation, exploratory analysis, model training, evaluation, and Grad-CAM-based explanation of predictions. The project compares conventional CNN architectures with transformer-based methods to understand both classification accuracy and interpretability.

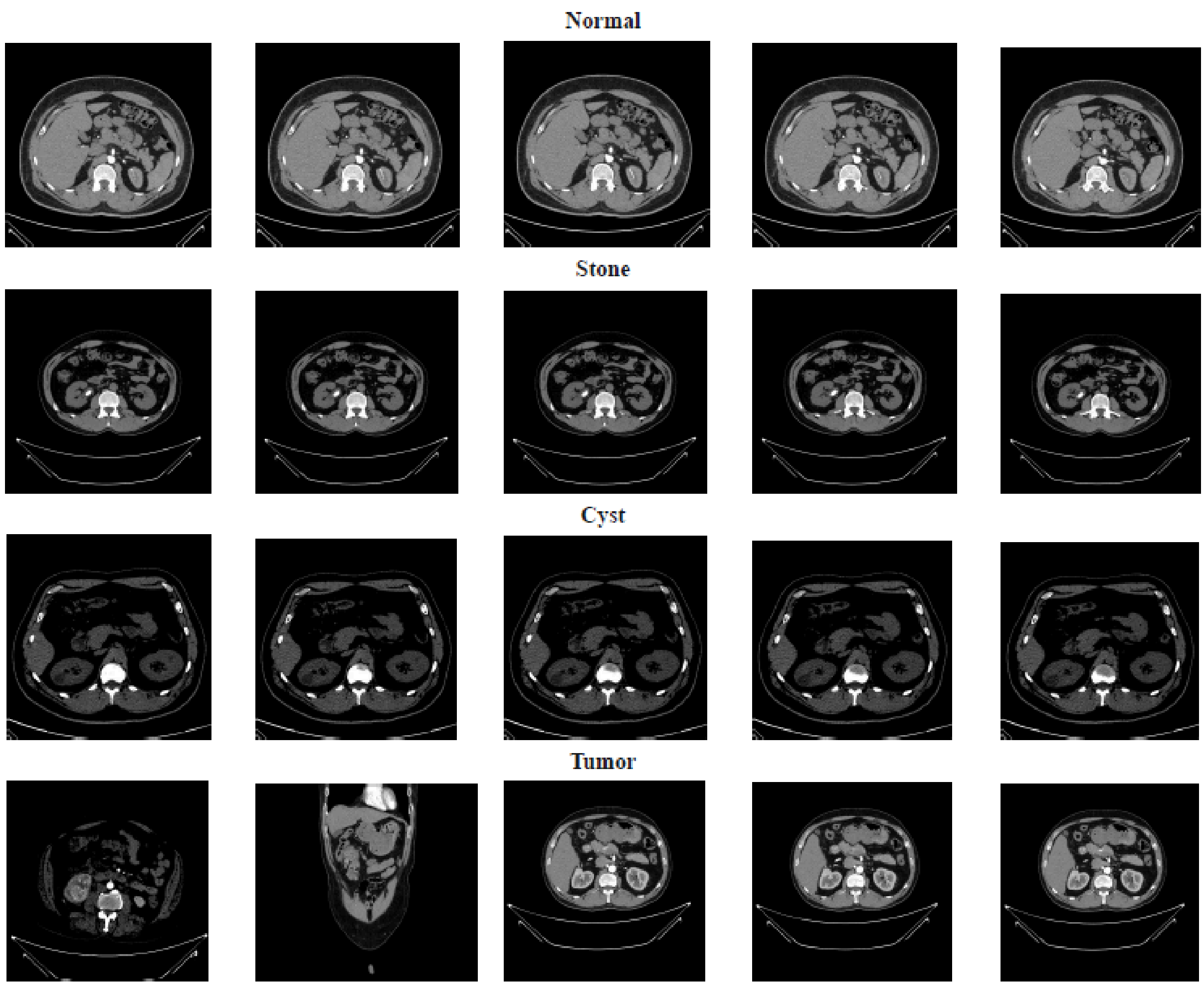
4.1 Methodology (Dataset, Preprocessing, Models)

This project uses the "CT KIDNEY DATASET: Normal–Cyst–Tumor and Stone", which contains 12,446 annotated CT images of the abdomen and urograms categorized into four classes: normal kidney, cyst, stone, and tumor. The images were originally collected from hospital PACS systems and workstations, anonymized, and converted from DICOM to standard image formats while removing all patient-identifiable information. Each image was carefully reviewed and labeled by medical experts to ensure that the ground-truth class for every sample is reliable.

The preprocessing pipeline includes directory-based data collection, construction of a master DataFrame, and several quality checks. Image paths and labels are first read from class-wise folders (Cyst, Normal, Stone, Tumor) and stored in a tabular structure for easier manipulation. Duplicate entries are detected and removed to avoid data leakage between training and testing, and missing paths or labels are checked to prevent runtime errors during model training.

After cleaning, images are resized and normalized to a consistent resolution and intensity range suitable for deep learning models. The dataset is then split in a stratified way into training, validation, and test sets so that the proportion of each class remains similar across all splits. Label encoding converts categorical labels (Cyst, Normal, Stone, Tumor) into integer values, which are required by the chosen loss functions and training routines.

Six deep learning architectures are implemented on this dataset: three convolution-based models (VGG16, ResNet50, and a modified Vision Transformer head) and three transformer-based or attention-enhanced models (EANet, Swin Transformer, and DINO). The final layers of these models are adapted to the four-class setting, and transfer learning is used where appropriate to benefit from pretrained weights. For explainability, DINO is integrated with Grad-CAM to generate heatmaps that highlight discriminative regions in the CT images, helping to interpret the model's decisions from a clinical perspective.



4.2 Testing / Verification Plan

The verification strategy uses a separate test set that is never seen during training or hyperparameter tuning. After model training on the training split and monitoring on the validation split, final evaluation is carried out exclusively on this held-out test data to estimate real-world performance. Stratified splitting ensures that all four classes are represented fairly in each split, preventing biased evaluation towards majority classes.

Model performance is quantified using standard classification metrics: accuracy, precision, recall, F1-score, and confusion matrices. Accuracy provides an overall measure of correct predictions, while precision and recall highlight how well the models detect each specific class, which is critical when misclassification of a tumor or stone can have serious clinical implications. Confusion matrices are analyzed to understand typical misclassification patterns, such as confusion between cyst and tumor, and to compare how different models handle difficult cases.

In addition to numeric metrics, the pipeline includes sanity checks for the data and training process. These checks confirm that there is no overlap between training and test images, that labels are consistent after encoding, and that the input image dimensions match model expectations. Grad-CAM visualizations also act as a qualitative verification step: if the highlighted regions consistently fall on kidneys and lesions rather than irrelevant background areas, the model's behavior is considered more trustworthy.

Subset	Proportion	Purpose	Shuffle
Training	80%	Model training	Yes
Validation	10%	Hyperparameter tuning	Yes
Test	10%	Final evaluation	No

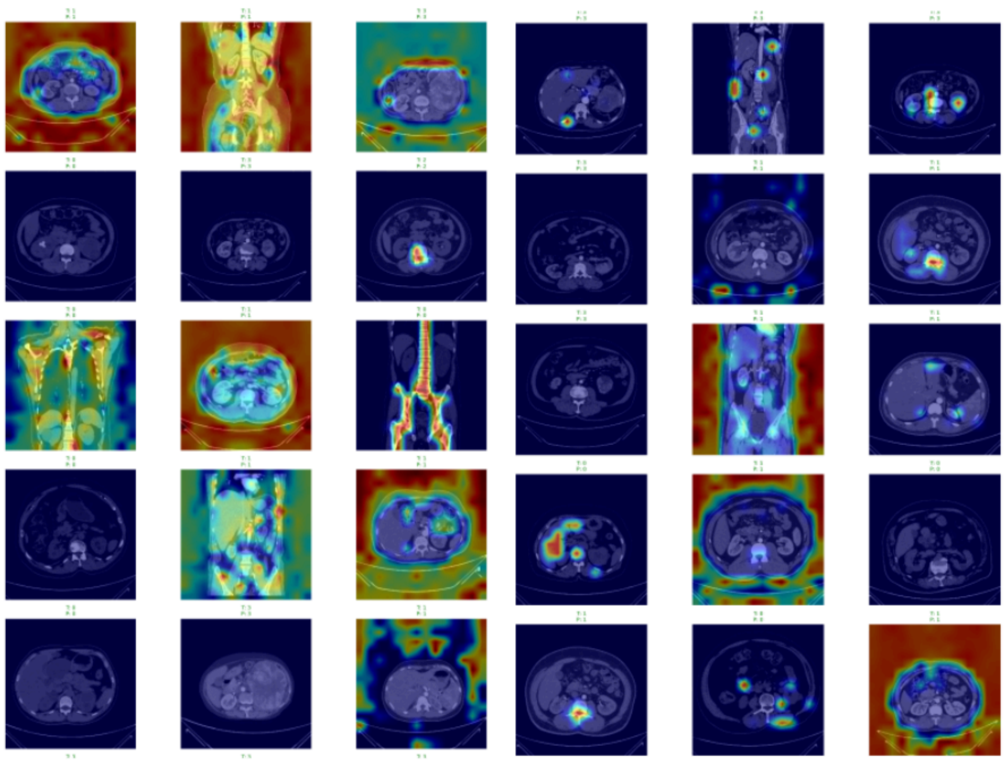
4.3 Result Analysis

The experimental results show that transformer-based models outperform conventional CNN architectures on this multi-class kidney CT classification task. Among all the tested models, the Vision Transformer (ViT) achieves the best performance with an accuracy of 99.37 percent on the prepared dataset. Comparative analysis of F1-score, precision, and recall confirms that ViT consistently provides superior detection across the four classes: normal, cyst, stone, and tumor.

The study also observes that DINO and VGG16 produce more informative Grad-CAM visualizations, offering clearer localization of pathological areas such as stones and tumors on CT slices. In contrast, ResNet50 shows comparatively lower performance in both classification metrics and localization quality. This suggests that transformer-based and attention-enhanced architectures are better suited to capturing the global contextual information needed to distinguish subtle differences between cysts, stones, and tumors in CT images.

Overall, the result analysis indicates that combining high-capacity transformer models with a carefully curated dataset and robust preprocessing can deliver highly accurate and clinically meaningful predictions. The excellent numeric performance of ViT, together with the interpretability achieved via Grad-CAM for DINO and VGG16, supports the viability of the proposed framework as a decision-support tool for kidney disease diagnosis.

?



4.4 Quality Assurance

Quality assurance in this project begins with data quality verification, including duplicate detection, missing-value inspection, and visualization of label distributions. Duplicate images are removed before splitting the dataset to avoid data leakage, while missing paths or labels are identified and corrected or excluded from the pipeline. Class distribution plots (bar and pie charts) are used to monitor imbalance among the four classes and to motivate the use of stratified splitting and other mitigation strategies.

The training pipeline is designed for robustness and reproducibility by using a central DataFrame as the single source of truth for image paths and labels, and by applying consistent label encoding prior to all model experiments. Standard practices such as fixed random seeds, clear separation of training, validation, and test sets, and standardized evaluation metrics further strengthen the reliability of results. From a clinical perspective, the integration of Grad-CAM-based explainability provides an additional layer of quality assurance, since visual inspection of heatmaps allows experts to verify whether model attention aligns with medically relevant kidney structures and lesions.

TABLE V: Measures of performance for the six models studied in the research.

Models	Accuracy	Class	Precision (PPV)	Recall (Sensitivity)	F1 Score
EANet [21]	98.35%	Cyst	0.96	0.98	0.97
		Normal	0.99	0.98	0.98
		Stone	1.00	0.84	0.91
		Tumor	0.92	1.00	0.96
Swin Transformer [19]	97.27%	Cyst	0.94	0.99	0.97
		Normal	1.00	0.99	0.99
		Stone	0.98	0.86	0.92
		Tumor	0.96	0.98	0.97
DINO	98.00%	Cyst	1.00	0.99	0.99
		Normal	1.00	0.95	0.98
		Stone	0.96	1.00	0.98
		Tumor	0.92	1.00	0.96
VGG16	98.47%	Cyst	0.98	0.99	0.98
		Normal	0.97	0.99	0.98
		Stone	0.98	0.97	0.97
		Tumor	1.00	0.95	0.97
		Cyst	1.00	1.00	1.00

$$\mathrm{Accuracy}_i = \frac{\mathrm{TP}_i + \mathrm{TN}_i}{\mathrm{TP}_i + \mathrm{TN}_i + \mathrm{FP}_i + \mathrm{FN}_i} \times 100\% \tag{4}$$

$$\mathrm{Precision}_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i} \tag{5}$$

$$\mathrm{Sensitivity}_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FN}_i} \tag{6}$$

$$\mathrm{F1\_score}_i = 2 \times \frac{\mathrm{Precision}_i \times \mathrm{Sensitivity}_i}{\mathrm{Precision}_i + \mathrm{Sensitivity}_i} \tag{7}$$

Chapter 5
Standards Adopted

5.1 Design Standards

The design of this project follows a structured AI pipeline that is commonly adopted in medical imaging research: data acquisition, preprocessing, model development, evaluation, and result interpretation. The system is organized as a modular architecture where each stage (dataset handling, model training, and Grad-CAM–based explainability) is clearly separated, which aligns with standard software and research design practices. The report itself is prepared according to the KIIT project report format and is conceptually consistent with IEEE-style research documentation used in the reference paper.

5.2 Coding Standards

The implementation adheres to basic coding standards used in data science and deep learning workflows. The dataset is handled through well-defined data structures (such as Pandas DataFrames) to ensure clear separation between file paths, labels, and derived attributes. The training code follows a modular structure in which data loading, preprocessing, model definition, training, and evaluation are implemented as distinct steps, enabling easier debugging and reproducibility. Standard libraries such as TensorFlow/Keras and scikit-learn are used as they provide stable, community-tested APIs for model training, optimization, and metric calculation.

5.3 Testing Standards

Testing in this project follows widely accepted machine learning evaluation standards. The dataset is split into training, validation, and test sets, ensuring that the final performance is reported only on previously unseen data. Classification performance is assessed using accuracy, precision, recall, F1 score, and confusion matrix, which are standard metrics for supervised multi-class classification problems. In addition, Grad-CAM visualizations are used as an interpretability check to verify that the model focuses on clinically meaningful regions in CT images when predicting cyst, stone, tumor, or normal cases.

Chapter 6

Conclusion and Future Scope

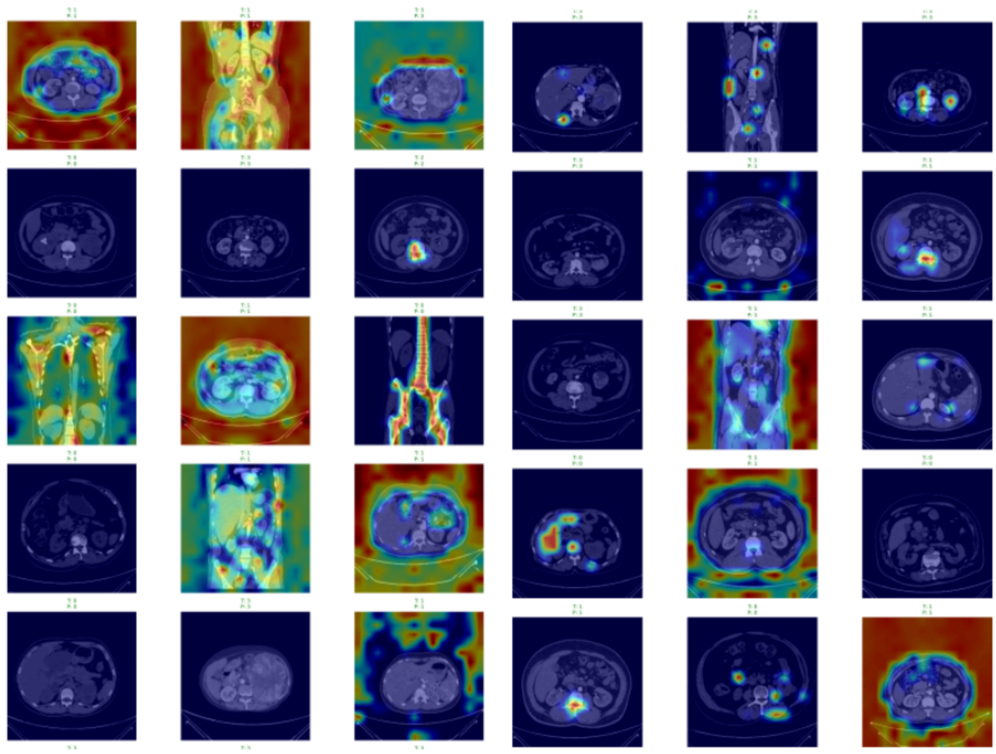
6.1 Conclusion

This project presented an AI-based framework for automatic multi-class classification of kidney CT images into normal, cyst, stone and tumor categories. Using the CT KIDNEY DATASET with 12,446 annotated images, six deep learning models were implemented, including VGG16, ResNet50, Vision Transformer (ViT), Swin Transformer, EANet and DINO. Experimental results showed that the Vision Transformer achieved the best performance with an accuracy of 99.37%, while transformer-based approaches in general outperformed conventional CNN architectures on this task. Grad-CAM visualizations using models such as DINO and VGG16 further demonstrated that the network focuses on clinically relevant regions, improving interpretability for medical users. Overall, the work indicates that transformer-based models combined with explainability techniques can substantially support radiologists and nephrologists in diagnosing kidney stones, cysts and tumors from CT scans.

6.2 Future Scope

Future work can extend this framework by validating the models on larger and more diverse multi-center datasets to ensure robust performance across different scanners, imaging protocols and patient populations. In particular, external validation on real clinical data from multiple hospitals would help assess generalizability and uncover potential failure modes in under-represented subgroups. The system can also be integrated into clinical workflows as a decision-support tool that provides both predicted labels and attention maps at the point of care, for example within PACS or hospital information systems.

Further research may explore model optimization for real-time inference on standard clinical hardware, improved handling of rare or borderline cases through techniques such as data augmentation or few-shot learning, and extension of the pipeline to additional renal or abdominal pathologies. Finally, this approach can contribute towards building digital-twin representations of renal function, enabling more comprehensive AI-assisted monitoring, risk stratification and treatment planning for patients with chronic kidney disease over time.



References

[1]S. M. Metev and V. P. Veiko, Laser Assisted Microtechnology, 2nd ed., R. M. Osgood, Jr., Ed. Berlin, Germany: Springer-Verlag, 1998.

Breckling, Ed., The Analysis of Directional Time Series: Applications to Wind Speed and Direction, ser. Lecture Notes in Statistics. Berlin, Germany: Springer, 1989, vol. 61.

S. Zhang, C. Zhu, J. K. O. Sin, and P. K. T. Mok, "A novel ultrathin elevated channel low-temperature poly-Si TFT," IEEE Electron Device Lett., vol. 20, pp. 569–571, Nov. 1999.

M. Wegmuller, J. P. von der Weid, P. Oberson, and N. Gisin, "High resolution fiber distributed measurements with coherent OFDR," in Proc. ECOC'00, 2000, paper 11.3.4, p. 109.

R. E. Sorace, V. S. Reinhardt, and S. A. Vaughn, "High-speed digital-to-RF converter," U.S. Patent 5 668 842, Sept. 16, 1997.

(2002) The IEEE website. [Online]. Available: http://www.ieee.org/

[7] M. Shell. (2002) IEEEtran homepage on CTAN. [Online]. Available: http://www.ctan.org/tex-archive/macros/latex/contrib/supported/IEEEtran/

INDIVIDUAL CONTRIBUTION REPORT:

TITLE: AI-Based Classification of Kidney Diseases Using Deep Learning Models

SHIRSH MOHAN
Roll No: 2305245

Abstract:
The project focuses on developing an AI-based system for classification of kidney diseases (cyst, tumor, stone, normal) using CT scan images. Multiple deep learning models including CNNs and transformer-based architectures were implemented. The aim is to improve diagnostic accuracy and assist medical professionals through automated classification.

Individual contribution and findings:
I was primarily responsible for **data preprocessing and implementation of the Vision Transformer (ViT) model**. I performed dataset cleaning, duplicate removal, and missing value handling to ensure data integrity. I implemented label encoding and stratified data splitting to maintain class distribution.

I handled image preprocessing using normalization, resizing (224×224), and dataset structuring using DataFrames. I also applied **random oversampling** to address class imbalance.

On the modeling side, I implemented and trained the **Vision Transformer (ViT)** using transfer learning. I tuned hyperparameters such as learning rate, batch size, and optimizer (Adam). My findings showed that ViT achieved the **highest accuracy (~99%)**, outperforming CNN-based models.

I gained strong experience in:

- Handling real-world medical datasets
- Transformer architectures
- Training stability and convergence

Full Signature of Supervisor: Full signature of the student:

…………………………… Shirsh Mohan ………………………………

INDIVIDUAL CONTRIBUTION REPORT:

SAURABH SHARMA
Roll No: 2305076

Abstract:
The project focuses on designing an AI-driven system for kidney disease classification using CT images, combining multiple deep learning architectures and preprocessing techniques.

Individual contribution and findings:
I worked on the **research and overall project design**. I conducted a detailed literature review on:

- Deep learning in medical imaging
- CNN vs Transformer architectures
- Existing kidney disease classification methods

I helped in:

- Designing the **complete project pipeline**
- Selecting appropriate models
- Understanding dataset requirements

My findings:

- Transformer models outperform CNNs in capturing global features
- Lack of multi-class datasets is a key research gap
- Explainability is essential in healthcare AI

Individual contribution to project report preparation:
I contributed to:

- **Introduction**
- **Background Study**
- Literature review

Individual contribution for project presentation and demonstration:
I presented:

- Research motivation
- Problem statement
- Existing solutions vs our approach

Full Signature of Supervisor: Full signature of the student:

…………………………… ……………………………

INDIVIDUAL CONTRIBUTION REPORT:

SOHAM DEY
Roll No: 2305250

Abstract:
The project presents an AI-based approach for kidney disease classification using CT scan images and multiple deep learning models to improve diagnostic accuracy.

Individual contribution and findings:
I worked on **data collection, dataset organization, and overall integration of the project**.

My responsibilities included:

- Managing dataset structure
- Organizing CT scan images into labeled directories
- Assisting in dataset annotation and validation

I also coordinated:

- Integration of preprocessing + models
- Testing complete pipeline
- Ensuring smooth execution of experiments

Findings:

- Proper dataset organization is critical for model performance
- Integration challenges helped understand end-to-end ML pipelines
- Collaboration is key in large AI projects

Individual contribution for project presentation and demonstration:
I handled:

- Project flow explanation
- Dataset overview
- End-to-end pipeline demonstration
-

Full Signature of Supervisor: ……………………………

Full signature of the student: Soham Dey ……………………………………

INDIVIDUAL CONTRIBUTION REPORT:

TITLE: AI-Based Classification of Kidney Diseases Using Deep Learning Models

SATWIK GUPTA
Roll No: 2305075

Abstract:
The project focuses on developing an AI-based system for classification of kidney diseases (cyst, tumor, stone, normal) using CT scan images. Multiple deep learning models including CNNs and transformer-based architectures were implemented. The aim is to improve diagnostic accuracy and assist medical professionals through automated classification.

Individual contribution and findings:
I was primarily responsible for **data preprocessing and implementation of the Vision Transformer (ViT) model**. I performed dataset cleaning, duplicate removal, and missing value handling to ensure data integrity. I implemented label encoding and stratified data splitting to maintain class distribution.

I handled image preprocessing using normalization, resizing (224×224), and dataset structuring using DataFrames. I also applied **random oversampling** to address class imbalance.

On the modeling side, I implemented and trained the **Vision Transformer (ViT)** using transfer learning. I tuned hyperparameters such as learning rate, batch size, and optimizer (Adam). My findings showed that ViT achieved the **highest accuracy (~99%)**, outperforming CNN-based models.

I gained strong experience in:

- Handling real-world medical datasets
- Transformer architectures
- Training stability and convergence

Full Signature of Supervisor: ……………………………

Full signature of the student: Satwik Gupta ……………………………..

INDIVIDUAL CONTRIBUTION REPORT:

TITLE: AI-Based Classification of Kidney Diseases Using Deep Learning Models

SWAYAM KASHYAP
Roll No: 2305258

Abstract:
This project aims to classify kidney diseases using deep learning models applied on CT scan images. Both CNN-based and transformer-based architectures were evaluated for performance comparison.

Individual contribution and findings:
I was responsible for implementing **CNN-based models (VGG16 and ResNet50)** using transfer learning.

- Modified final layers for multi-class classification
- Fine-tuned models to avoid overfitting
- Compared CNN performance with transformer models

Findings:

- **VGG16 performed well (~98%)** with good feature extraction
- **ResNet50 showed lower performance (~94%)**
- CNNs are effective but limited in capturing global dependencies

I also worked on model evaluation metrics such as:

- Accuracy
- Precision, Recall
- F1-score

Individual contribution for project presentation and demonstration:
I explained:

- CNN architecture differences
- Model comparison results
- Evaluation metrics

Full Signature of Supervisor: ....

Full signature of the student: Swayam Kashyap ....

TURNITIN PLAGIARISM REPORT

(This report is mandatory for all the projects and plagiarism must be below 25%)

Sample_turnitin_report_for_students.docx

ORIGINALITY REPORT

13%	7%	3%	7%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Kennesaw State University Student Paper	3%
2	www.guardian.co.uk Internet Source	2%
3	Campbell, Neil. "Post-Western Cinema", A Companion to the Literature and Culture of the American West Witschi/A Companion to the Literature and Culture of the American West, 2011. Publication	1%